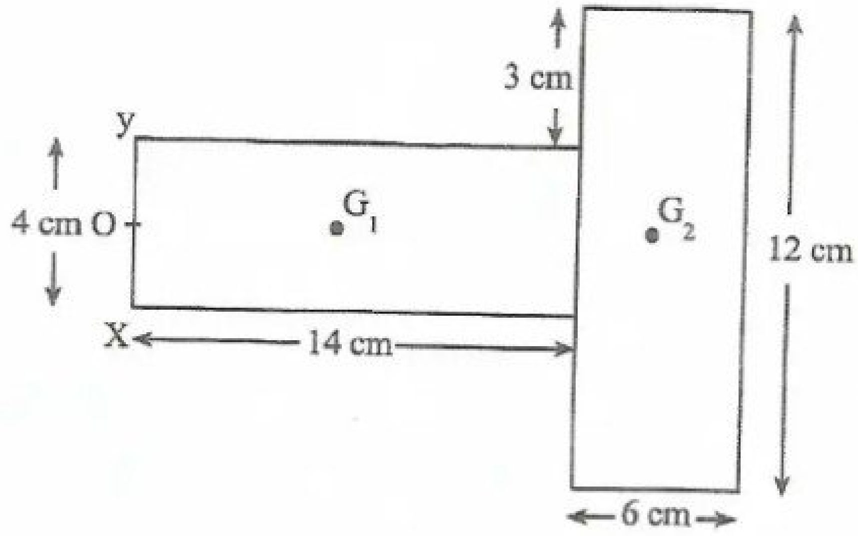
1




The diagram shows two uniform rectangular laminae, of mass per unit area $m$. The laminae are joined together.

Find the distance of the centre of gravity of the complete lamina from the edge XY.

[3]

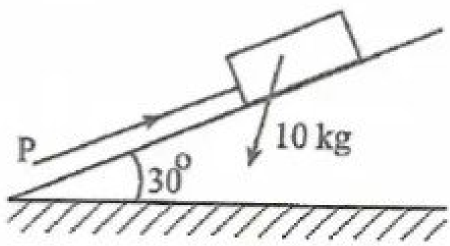
2 A car of mass 1 200 kg travels up a road inclined at $5°$ to the horizontal. The resistance to motion of the car is 325 N and it is moving with a speed of 8 $ms^{-1}$. The engine of the car is working at a rate of 15 kW.

Find for the car the

**(a)** driving force, [1]

**(b)** acceleration. [3]

3




The diagram shows a force $PN$ acting parallel to and up a line of greatest slope which holds a particle of mass 10 kg in limiting equilibrium on a smooth plane inclined at $30°$ to the horizontal.

Find the

**(a)** magnitude of P, [3]

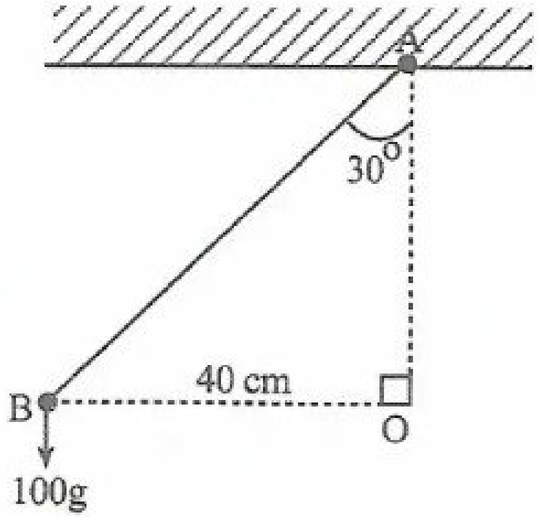
**(b)** normal reaction between the particle and the plane. [2]

4 A bead of mass 5 kg is threaded on to a smooth wire in the shape of a circle radius 3 m and centre O. The circular wire is fixed in a vertical plane and the bead passes through the lowest point A on the wire with speed $12\ ms^{-1}$. When the bead is at the top of the wire, find the

(a) velocity of the bead, [3]

(b) magnitude of the reaction between the bead and the wire. [2]

5




A light inextensible string fixed at A and a particle B of mass 100 grams is attached as shown in diagram. The particle moves in a horizontal circle of radius 40 cm and centre O, vertically below A. The angle between the string and the vertical is 30°.

Find the

(a) tension in the string, [3]

(b) angular speed of the particle. [3]

6 Three boys are pulling a heavy trolley by means of three ropes on horizontal ground. The boy in the middle is exerting a force of 80 N due North. The other two boys, whose ropes both make an angle of 30° with the centre rope, are pulling with forces of 60 N and 120 N.

Calculate the

(a) magnitude of the resultant of the pull on the trolley. [4]

(b) direction of the resultant of the pull on the trolley. [2]

7 A particle starts from rest and travels in a straight line with an acceleration $cos\pi t$, where $t$ is the time.

(a) Show that $v = \frac{I}{\pi} sin\pi t$. [3]

(b) Hence or otherwise, find the distance covered by the particle in the interval of time from $t = 2$ to $t = 3$. [3]

8 Two identical spheres P and Q of masses 250 grams and 550 grams, respectively are moving towards each other along the same horizontal line with speeds of 5 $ms^{-1}$ and 3 $ms^{-1}$ respectively. The particles collide directly and Q moves on with a speed of 1.5 $ms^{-1}$ while P moves in the opposite direction of its initial direction.

Find the

(a) speed of P, [3]

(b) loss of kinetic energy due to the collision. [3]

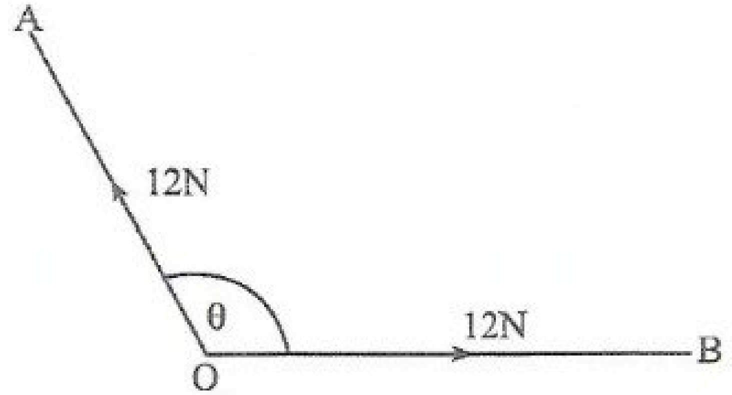
9 A particle of mass 0.4 kg is projected down a slope at a speed of 11 $ms^{-1}$. The slope is inclined at $sin^{-1}\left(\frac{6}{25}\right)$ to the horizontal. A resisting force P N acting on the particle reduces the speed of the particle to 1$ms^{-1}$ over a distance of 3 m.

Find the

(a) loss in potential energy of the particle, [3]

(b) value of P. [3]

10



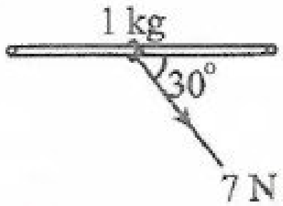

The diagram shows two forces, each of magnitude 12 **N**, act at a point O in the directions OA and OB. The angle between the forces is $\theta$. The resultant of these two forces in 14 **N**.

Find the

**(a)** obtuse angle $\theta$, [4]

**(b** component of the resultant force in the direction OA. [3]

11




The diagram shows a ring of mass 1 kg threaded on a rough rod which is fixed horizontally. The system is in limiting equilibrium when it is pulled by a force of magnitude 7 N at 30° below the horizontal (see diagram).

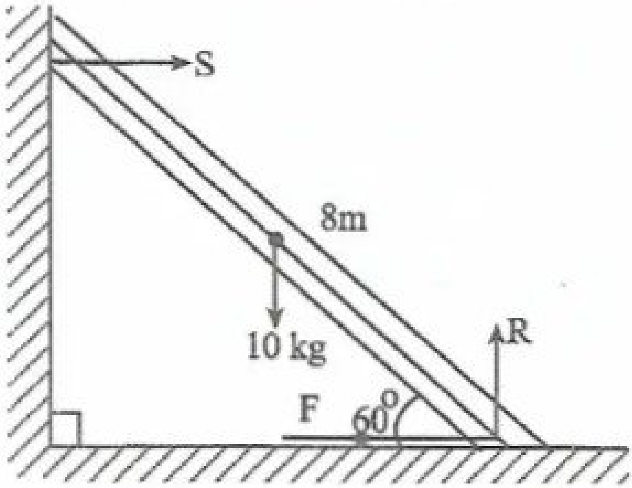
Find the

**(a)** normal reaction, [3]

**(b)** frictional force exerted by the rod on the ring, [2]

**(c)** co-efficient of friction between the rod and the ring. [2]

12




The diagram shows a uniform ladder of mass 10 kg and length 8 m resting in equilibrium at an angle of 60° to the horizontal. Its upper end is resting against the smooth vertical wall and its lower end on rough horizontal ground, the co-efficient of friction is $\mu$. The centre of gravity of the ladder is 4 m from the foot of the ladder.

Find the

**(a)** forces, S, F and R, [4]

**(b)** least possible value of $\mu$. [3]

13 Two cars A and B are 90 m apart on a smooth horizontal surface. A is moving directly towards B with speed 6 $ms^{-1}$ and retardation 1 $ms^{-2}$. B is moving directly towards A with speed 2 $ms^{-1}$ and acceleration 3 $ms^{-2}$.

**(a)** Find the displacement of car A and car B in terms of $t$, the time at any instant. [4]

**(b)** Hence or otherwise find the time taken before the two cars meet. [3]

14 A ball is projected with an initial velocity of 20 $ms^{-1}$ at an angle of 45° above the horizontal.

Find the

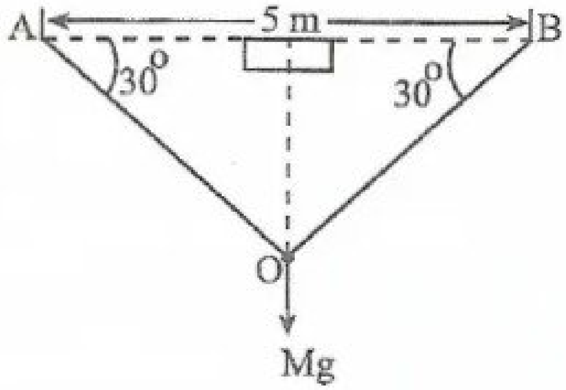
**(a)** exact speed of the ball 3 seconds after projection. [4]

**(b)** distance of the ball from its projection 1 second after being projected. [4]

15 **(a)** An elastic string of natural length 3 m is fixed at one end and is str to 4,8 in length by a force of 8 N.

Find the modulus of elasticity of the string. [3]

**(b)**




The diagram shows an elastic string of natural length 5 m and modulus of elasticity 5 g is stretched between two points A and B which are on the same level, where AB = 5 m. A particle of mass M is attached to the mid-point of the string and hangs in equilibrium with both parts of the string making angles of 30° with AB.

Find the

**(i)** extension of the string, [3]

**(ii)** mass $M$ of the particle. [2]

16 **(a)** The forces $\mathbf{F_1}$ and $\mathbf{F_2}$ are such that, $\mathbf{F_1} = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$, $\mathbf{F_2} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$.

Calculate

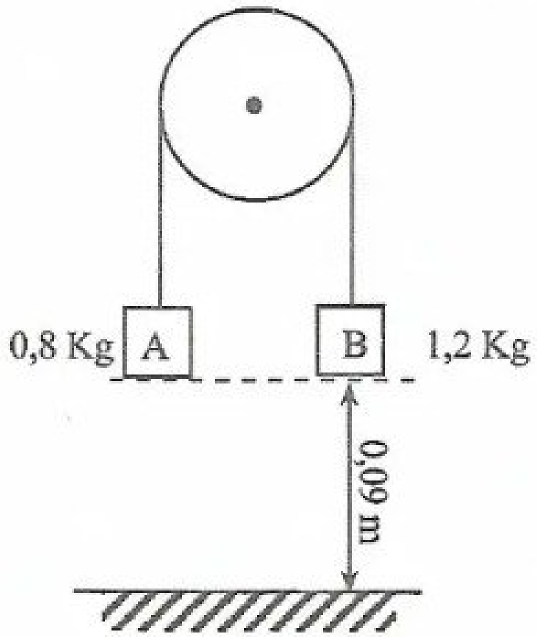
**(i)** $\mathbf{F_1} + \mathbf{F_2}$, [1]

**(ii)** $|\mathbf{F_1} + \mathbf{F_2}|$. [2]

**(b)** **(i)** Find the vector equation of the line passing through the point with position vector $-4\boldsymbol{i} - \boldsymbol{j} + \boldsymbol{k}$, which is parallel to the line with vector equation $\boldsymbol{r} = 2\boldsymbol{i} + 3\boldsymbol{j} + \lambda(\boldsymbol{i} + \boldsymbol{j} - \boldsymbol{k})$. [2]

**(ii)** Determine the point of intersection of the following pairs of lines:
$r_1 = \boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k} + t(-2\boldsymbol{i} + \boldsymbol{j} - 3\boldsymbol{k})$ and
$r_2 = \boldsymbol{j} - 3\boldsymbol{k} + s(\boldsymbol{i} - \boldsymbol{j} + \boldsymbol{k})$. [4]

**17**




The diagram above shows two particles A and B, connected by a light inextensible string which passes over a smooth fixed pulley. The system is held with the string taut and with A and B each at a height of 0,09 m above a fixed horizontal plane. It is then released from rest. When B reaches the plane it becomes stationary.

**(a)** Find

**(i)** acceleration of the system while both particles are in motion, [4]

**(ii)** speed with which B hits the plane, [2]

**(b)** Calculate the maximum height above the plane attained by A. [3]

**18** **(a)** Define simple harmonic motion. [2]

**(b)** A particle is projected from a point O at time $t = 0$ and performs simple harmonic motion with O as the centre of oscillation. The motion is of amplitude 20 cm and time period 4 $s$.

Find the

**(i)** maximum speed of projection, [3]

**(ii)** speed of the particle when $t = 1.5$ $s$. [3]

**(iii)** value of $t$ when the particle is first at a point 10 cm from 0. [3]